

DECODELABS INTERSHIP 2026

WEEK 2 - PROJECT 2: EXPLORATORY DATA ANALYSIS (EDA)

REPORT

- **Intern Name:** Martina Marco Dawoud
- **Track:** Data Analytics
- **Project Title:** Exploratory Data Analysis on Transaction Dataset

1. EXECUTIVE SUMMARY

The primary objective of this project is to perform an Exploratory Data Analysis (EDA) on the cleaned transaction dataset. By applying descriptive statistics and evaluating core metrics, this analysis uncovers hidden transaction patterns, identifies structural anomalies (outliers), and defines critical business Key Performance Indicators (KPIs) to drive enterprise intelligence.

2. DESCRIPTIVE STATISTICS TABLE

The table below summarizes the central tendency and distribution metrics for the primary numerical attributes within the dataset: Quantity, UnitPrice and TotalPrice.

Statistical Metric	Quantity	UnitPrice	TotalPrice
Mean	2.95	356.41	1053.97
Median	3.00	364.21	823.62
Mode	1.00	127.18	635.90
Minimum Value	5.00	699.93	3456.40
Maximum Value	1.00	11.39	11.39

3. CENTRAL TENDENCY & DISTRIBUTION ANALYSIS

- **Skewness Metrics:** For the core financial metric TotalPrice, the calculated **Mean (\$1,053.97)** is notably higher than the **Median (\$823.62)**. This specific mathematical relationship confirms that the transactional data distribution is **positively skewed (right-skewed)**.
- **Interpretation:** The right-hand skewness proves that while the baseline volume of customer orders is concentrated around lower to middle price points (under the median), a smaller tier of high-value transactions effectively shifts the mathematical average upward.

4. OUTLIER DETECTION & INSIGHTS

- **Anomalies Identified:** Transactions reaching the upper bound ceiling near the **Maximum Value (\$3,456.40)** represent the absolute outliers within our distribution.
- **Underlying Logic:** Verification shows these outliers are driven by wholesale purchase behaviors, combining the maximum transactional capacity (Quantity = 5) with premium inventory segments where the UnitPrice approaches its highest tier (\$699.93). These entries are verified as legitimate business spikes rather than systemic errors.

5. CORE BUSINESS KEY PERFORMANCE INDICATORS (KPIs)

Based on the absolute totals and calculations within the optimized dataset, the foundational financial and operational baselines are defined as follows:

1. **Total Gross Revenue: \$1,257,384.18** *(The cumulative gross value generated across all recorded baseline sales transactions).*
2. **Total Units Sold: 3,514 Units** *(The total volume of inventory items successfully moved and distributed).*
3. **Average Order Value (AOV): \$1,053.97** *(The mean financial size per customer basket, showcasing healthy individual order performance).*

6. ADVANCED IMPLEMENTATION: NET REVENUE AUDIT

- **Strategic Distinction:** To refine administrative visibility, a custom operational logic was introduced to differentiate Gross financial metrics from actual Net yield.
- **The Net-TotalProfit Column:** A conditional logical script (IFS) was implemented across the dataset to audit active marketing coupon impacts (such as deducting 10% for SAVE10 and 15% for WINTER15). This ensures that promotional discounts are mathematically isolated, allowing the business to pinpoint true net margins against gross processing totals.

7. CONCLUSION

This exploratory phase successfully translates raw, isolated rows into solid enterprise insights. The variance between Mean and Median structures highlights a steady customer baseline with occasional high-yield orders, providing a mathematically sound foundation for the upcoming predictive visualization and dashboard phase.